

A Bio-Inspired Jump-Gliding Robot for Planetary and Terrestrial Exploration

Final Technical Report

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Abstract—Inspired by recent studies of planetary jump robots and bio-inspired gliding mechanisms, we propose a novel robot capable of both repeated elastic-powered jumps and controlled gliding. The key innovation is the use of carbon fiber strips for energy storage and actuation, enabling jumping launches and mid-air wing deployment. So far, we detail the system architecture, mathematical modelling, and challenges encountered in the realization of this hybrid jump-gliding robot.

Index Terms—Jump-gliding robot; elastic-powered jumping; carbon fiber; gliding mechanism.

I. INTRODUCTION

Our main goal was to design a robot capable of repeated jumps and gliding, expanding its operational range and adaptability. Our innovation lies in the integration of a carbon fiber strip-powered jumping mechanism with a deployable wing gliding system. Unlike prior robots that typically perform single-use or limited jumps, and compared to conventional flying robots, our design offers higher reliability and significantly lower energy consumption. So far, we detail the system architecture, mathematical modeling, and challenges encountered in the realization of this hybrid jump-gliding robot.

II. SYSTEM DESIGN

Solution Selection: After comparing different jump actuation mechanisms, we found that biomimetic legged or complex multi-link jumping structures are not suitable for our intended application. These mechanisms are generally too heavy and not conducive to achieving lightweight design and effective gliding. Based on further research, we selected the carbon fiber strip compression jumping mechanism, which best meets our requirements for both performance and weight.

Release Mechanism: The release mechanism works by using an electromagnetic clutch* to quickly connect or disconnect the winch from the motor. During energy storage, the motor—through the engaged clutch—pulls the UHMWPE line to stretch the carbon fiber strip and store elastic energy. When the target tension is reached, power to the clutch is cut, instantly disconnecting the winch from the motor. This allows the stored energy to be released rapidly, enabling the jump, while

protecting the motor from any sudden forces or backlash during the release.

Design of Jump Energy Storage and Release: Utilizes a winch motor to tension an Ultra-high-molecular-weight polyethylene (UHMWPE) line attached to carbon fiber strips, storing elastic potential energy.

Electromagnetic Clutch

For the release mechanism, we employed an electromagnetic clutch to control the tightening and release of the Ultra-high-molecular-weight polyethylene (UHMWPE). To ensure proper alignment of the UHMWPE during retraction, a dedicated holder frame (Fig. 1) was designed to accommodate both the servo motor and the electromagnetic clutch.



Fig. 1. Holder frame

In addition, a secondary frame was introduced to support the winch shaft, enabling it to counteract the downward tensile force exerted by the rope. Overall, the servo motor is rigidly connected to the electromagnetic clutch via a coupling component (Fig. 2.), and both are mounted within the frame. The gear portion of the clutch is exposed externally (Fig. 3.), where it engages with a custom-designed internal-toothed shaft connector. This shaft passes through a bearing, which is secured on the secondary frame (Fig. 4.). The shaft itself is designed with a small through-hole (Fig. 5.), through which the UHMWPE is threaded and fixed by knotting, enabling controlled winding and unwinding.

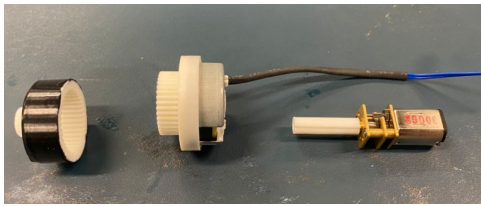


Fig. 2. Servo-clutch coupling

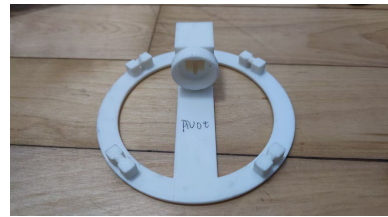


Fig. 6. First version of upper plate design



Fig. 3. Clutch placement



Fig. 4. Bearing support



Fig. 5. Shaft with through-hole

The design underwent several iterations before reaching the final version:

First Version (Fig. 6.): The initial frame did not adequately secure the servo motor and clutch. During testing, once the UHMWPE was tensioned to a certain degree, the shaft connector would detach from the frame. Although adhesive was applied to fix the clutch, the shaft of the connector lacked sufficient strength and eventually fractured under load.

Second Version (Fig. 7a, b): The frame dimensions were revised, and a larger supporting frame was added to house the bearing. However, this resulted in a bulkier structure, and the shaft connector continued to experience detachment.



Fig. 7. a Second version

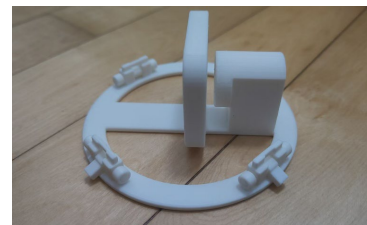


Fig. 7. b. Second version

Third Version: The geometry of the winch frame was further optimized by adjusting both the frame spacing and shaft dimensions of the connector. In this iteration, the clutch and bearing were reliably secured, preventing detachment. The mechanism successfully supported the complete jumping process and enabled repeatable actuation.

Gliding Mechanism

Design: The gliding wings are constructed using multiple carbon fiber rods as the structural framework, over which a high-strength polymer kite fabric is stitched to provide lift during gliding. A dedicated wing mounting frame was also designed to secure the wings to the upper plate of the robot.

Preliminary Concept: In the initial configuration, the wings remain folded in a retracted state, hanging naturally downward. Upon reaching the apex of the jump and during the onset of descent, the wings deploy passively under aerodynamic reaction forces, thereby initiating gliding.

Design of Gliding Mechanism: Wings are deployed at the apex of each jump, either by electronic control or mechanical triggering.

*An electromagnetic clutch engages or disengages power transmission between two shafts by using electromagnetic force. When current is applied, the clutch connects the shafts; when current is removed, the connection is released.

III. MODELING AND ANALYSIS

Energy Storage:

$$E_{\text{elastic}} = \int_0^{x_{\text{max}}} F(x) dx$$

The above equation represents the idealized calculation of the elastic energy stored in our device, obtained by directly summing the discrete force measurements without considering energy losses. However, we found that results calculated using this method differ significantly from those obtained by the area-under-the-curve (integral) approach. In practical applications, it is more accurate to estimate the stored energy using the integral formulation shown below.

$$E_{\text{elastic}} \approx \sum_{i=1}^{n-1} \frac{F(i) + F(i+1)}{2} (x_{i+1} - x_i)$$

We define the stored elastic energy, E_{elastic} , as the total mechanical work done during the compression of the carbon fiber spring. Specifically, the force-displacement relationship is nonlinear, and the energy is computed by integrating the measured force $F(x)$ over the full compression distance x_{max} , where $F(x)$ is the measured force at displacement x , and x_{max} is the maximum compression. * The area under the force-displacement curve was obtained experimentally using a digital scale and ruler (see photos below), yielding an estimated elastic energy of approximately 15.2 J as shown in Fig. 8.

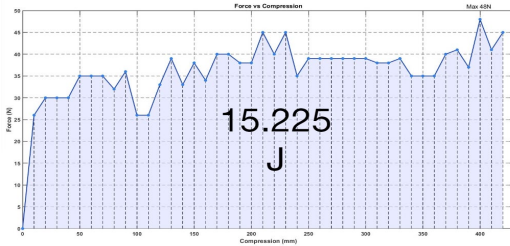


Fig. 8. Experimental force-displacement curve of the carbon-fiber spring

Theoretical Jump Height:

$$H_{\text{ideal}} = \frac{E_{\text{elastic}}}{M_{\text{total}} \cdot g}$$

The ideal jump height, H_{ideal} , is estimated by equating the measured elastic potential energy to the gravitational potential energy required to elevate the total system mass. Specifically, where E_{elastic} is the experimentally measured elastic energy, M_{total} is the total mass of the jumping system, and g is the acceleration due to gravity. This equation assumes 100% conversion of stored elastic energy into vertical displacement,

thus representing the upper bound of achievable jump height in the absence of energy losses.

Jump Height:

$$H_{\text{estimate}} = \frac{E_{\text{elastic}} \cdot (1 - D_{\text{drag}}) \cdot (1 - \beta_{xy} - \beta_{\theta}) \cdot (1 - x_{\text{CoM}})}{M_{\text{total}} \cdot g}$$

The jump height, H_{estimate} , accounts for various energy losses in the real system that are not reflected in the theoretical calculation. These losses include air drag (D_{drag}), effects from non-vertical take-off angles (β_{xy}), device rotation during take-off (β_{θ}), and the influence of the center of mass position (X_{CoM}).

where:

D_{drag} is the fractional energy loss due to air resistance during the jump

β_{xy} and β_{θ} represent losses caused by non-vertical take-off and device rotation, respectively

X_{CoM} denotes the effect of the center of mass location on the device

$$H_{\text{estimate}} = H_{\text{ideal}} \cdot \eta$$

As it is difficult to measure or model all these loss factors accurately, we also employ a simplified approach by introducing an overall efficiency factor η , defined as the ratio of actual jump height to the theoretical value based on literature [12].

* For energy estimation, a force gauge was attached to the jumping device and incrementally stretched in the vertical direction. The force readings were visualized and recorded at each interval to generate a force-displacement curve. By calculating the area under this curve using the appropriate formula (numerical integration), we obtained an estimate of the stored elastic energy in the system.

IV. PROTOTYPE AND EXPERIMENTS

This section presents the preliminary prototype we fabricated based on our prior design and simulations, along with the initial experimental tests conducted on several of the model's functional components.

A. Jumping Module

In the final prototype, four carbon - fiber strips measuring 75 cm $\times$ 0.5 cm $\times$ 0.2 cm were employed as the elastic energy - storage elements of the jumping mechanism. The strip ends were secured using carbon - fiber fabric and cloth - based adhesive tape, and a 1.4 m-long ultra-high-molecular-weight polyethylene (UHMWPE) braided line was routed through the assembly to preload the strips.

Structural platforms: The upper and a lower circular platform, each with a diameter of 130 mm and a thickness of 3.5 mm. These platforms are connected by custom joints and 3D-printed connectors, which provide the structural support for the motor, clutch, and jumping mechanism. The lower platform also anchors the UHMWPE tensioning lines required for energy storage.

Electrical module: The actuation is driven by a 1:1000 high-torque DC motor, controlled by an Arduino Nano microcontroller. Power is supplied by eight 100 mAh lithium-

polymer battery packs. The module also includes custom wiring and connectors to coordinate motor actuation and release control.

Jumping and release mechanism: The redesigned energy storage system, combined with the clutch-bearing release module, ensures repeatable high-energy jumps rather than one-time actuation.

The entire device employs extensive 3D-printed components and custom connectors, achieving a balance of light weight and mechanical robustness. In operation, the robot first stores elastic potential energy in the carbon-fiber strips, then rapidly releases it to propel the body upward.

The overall structure measures approximately 757 mm in length and 130 mm in width, the fully assembled jumping module has a mass of 173 g.

the prototype is shown in Fig. 9.



Fig. 9. The jumping module in the uncharged state, and in the charged with instantaneous - release state

The operation of the jumping mechanism proceeds through four distinct phases, as illustrated in Fig. 10:

(a) Resting state: The carbon-fiber strips are straight and unstressed, with the UHMWPE line relaxed. (b) Energy storage: The motor-clutch assembly tensions the UHMWPE line, bending the carbon-fiber strips and storing elastic potential energy. (c) Release: The release mechanism disengages, enabling the stored energy to be rapidly converted into kinetic energy. (d) Take-off: The accumulated elastic energy propels the structure upward, lifting the module off the ground.

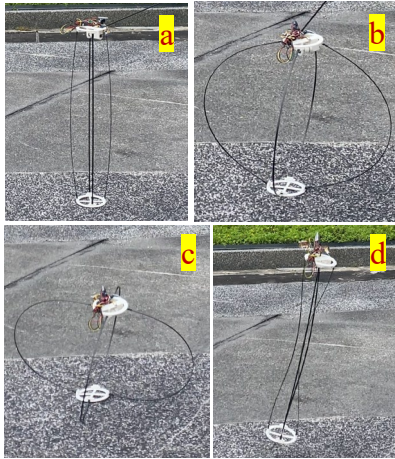


Fig. 10. Sequential operation of the jumping mechanism.

Experimental evaluation was conducted to verify the performance of the jumping module. The prototype achieved a maximum jump height of approximately 1.5 m, as shown in Fig. 11. Endurance testing demonstrated that the system can perform at least four consecutive jumps without structural damage. Under ideal conditions, the module is capable of repeated operation until the depletion of the onboard battery.



Fig. 11. Measured maximum jump height of the prototype (≈1.5 m)

A. Gliding Module

A preliminary gliding module was constructed to evaluate the feasibility of extending aerial motion following take-off. The structure employed carbon-fiber rods as the primary skeleton and was covered with a lightweight Icarex PC31 fabric, as shown in Fig. 12. This configuration provided a rigid yet lightweight wing frame suitable for integration with the jumping mechanism.

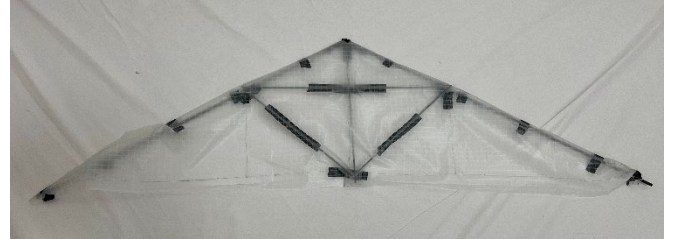


Fig. 12. Preliminary gliding module constructed using carbon-fiber rods and Icarex PC31 fabric

Independent testing of the gliding module indicated that the prototype was capable of achieving stable passive gliding when deployed in isolation, as shown in Fig. 13. Although the present tests were limited in scope, the results demonstrate the potential of combining the jump module with the glider to achieve controlled jump-gliding locomotion.

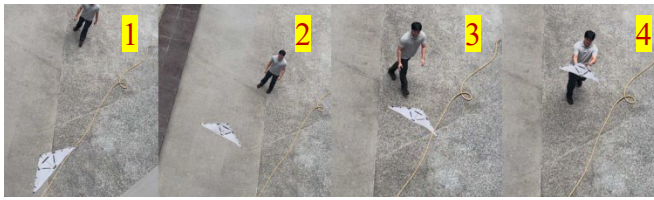


Fig. 13. Preliminary gliding module prototype constructed using carbon-fiber rods and Icarex PC31 fabric

V. CONCLUSION AND FUTURE WORK

In this work, we presented the design and prototyping of a bio-inspired jump-gliding robotic system. The proposed jumping mechanism, based on carbon-fiber elastic strips and a clutch-bearing release assembly, successfully achieved repeatable jumping performance, reaching heights of approximately 1.5 m. Unlike prior studies, which typically demonstrated only single-shot jumps due to irreversible structural deformation or limited energy-release mechanisms, our system has been experimentally validated to perform multiple consecutive jumps without structural failure. This marks a significant advancement in the domain of bio-inspired elastic jumping robots.

In parallel, a preliminary gliding module was developed using a carbon-fiber skeleton covered with Icarex PC31 fabric. Independent testing confirmed its capability to achieve passive gliding; however, full integration of the jumping and gliding subsystems has not yet been realized. The integration poses nontrivial challenges, including structural coupling, stability during transition, and control of aerodynamic surfaces.

Future work will focus on the mechanical and control integration of the jumping and gliding modules to achieve seamless jump-gliding locomotion. Additional efforts will also be directed toward optimizing the energy efficiency, enhancing structural robustness, and exploring control strategies that enable stable aerial transitions. Successful realization of such integration would open the door to practical applications in extraterrestrial exploration, search-and-rescue missions, and operations in extreme or hazardous environments.

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